

Query Quest Module — Problem Statement

Proxion 2026

A multinational analytics company maintains multiple relational databases containing employee records, customer transactions, product sales, project allocations, and platform activity logs. Your task is to solve advanced SQL challenges inspired by real-world industry scenarios and competitive programming platforms.

Participants must write optimized SQL queries for all tasks using only the provided datasets.

Important Submission Note

- Participants must submit all SQL queries in a single **PDF document** before the submission deadline.
 - The PDF must clearly mention the **question number** followed by its corresponding SQL query.
 - Any form of **AI-generated solutions or AI assistance during the competition is strictly prohibited.**
 - Participants found using AI tools, copied solutions, or unauthorized assistance will be **immediately disqualified.**
 - Teams must be able to explain the logic and working of every submitted query during evaluation or viva.
 - Queries should be properly formatted and readable to ensure smooth evaluation.
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Database Tables

Employees

Column

Type

employee_id	INT
employee_name	VARCHAR
department_id	INT
manager_id	INT
salary	INT
hire_date	DATE

Departments

Column	Type
department_id	INT
department_name	VARCHAR

Customers

Column	Type
customer_id	INT
customer_name	VARCHAR
city	VARCHAR

Orders

Column	Type
order_id	INT
customer_id	INT
order_date	DATETIME
amount	DECIMAL

Products

Column	Type
product_id	INT
product_name	VARCHAR
category	VARCHAR
price	DECIMAL

Sales

Column	Type
sale_id	INT
product_id	INT
employee_id	INT
quantity	INT
sale_date	DATE

Logins

Column	Type
user_id	INT
login_date	DATE

Logs

Column	Type
id	INT
num	INT

Weather

Column	Type
id	INT

recordDate	DATE
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temperature	INT
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OrderDetails

Column	Type
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order_id	INT
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product_id	INT
------------	-----

quantity	INT
----------	-----

SQL Challenges

Question 1

Tables Used:

- Employees

Find the second highest distinct salary from the Employees table.

Input:

employee_id	employee_name	salary
1	Ali	50000
2	Sara	70000
3	Ahmed	60000

Output:

SecondHighestSalary

60000

Question 2

Tables Used:

- Logins

Find all users who logged in for at least 3 consecutive days.

Example:

user_id	login_date
1	2026-01-01
1	2026-01-02
1	2026-01-03
2	2026-01-01

Output:

user_id
1

Question 3

Tables Used:

- Employees

Find duplicate employee names along with the number of occurrences.

Example

Input:

employee_name

Ali

Sara

Ali

Output:

employee_name	occurrences
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Ali	2
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Question 4

Tables Used:

- Employees

Retrieve managers who supervise at least 5 employees.

Example

Output: (Check out the Employees table for the solution's hint).

manager_id

Question 5

Tables Used:

- Employees
- Departments

Find employees who have the highest salary in their respective departments.

Example

Output:

department_name	employee_name	salary
IT	Sara	120000

Question 6

Tables Used:

- Customers
- Orders

Find customers who have never placed any order.

Example

Output:

customer_name

Hamza

Question 7

Tables Used:

- Weather

Find all dates where the temperature was higher than the previous day.

Example

Input:

recordDate	temperature
2026-01-01	20
2026-01-02	25

Output:

recordDate
2026-01-02

Question 8

Tables Used:

- Products
- Sales

Find products whose total sales quantity exceeds the average sales quantity of all products.

Example:

Output:

product_name

Laptop

Question 9

Tables Used:

- Customers
- Orders

Retrieve the customer with the highest total spending for each month.

Example

Output:

month customer_name total_spent

January	Sara	5000
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Question 10

Tables Used:

- Employees

Find employees whose salary is greater than their manager's salary.

Example

Output:

employee_name

Ahmed

Question 11

Tables Used:

- Logs

Find numbers that appear at least 3 times consecutively in the Logs table.

Example

Input:

id	num
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1	5
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2	5
---	---

3	5
---	---

4	2
---	---

Output:

ConsecutiveNums

5

Question 12

Tables Used:

- Sales

Calculate a running total of sales ordered by sale date.

Example

Output:

sale_date	running_total
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2026-01-01	10
------------	----

2026-01-02	25
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Question 13

Tables Used:

- Employees
- Departments

Rank employees based on salary department-wise using DENSE_RANK().

Example

Output:

employee_name	salary	rank
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Sara	100000	1
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Ali	90000	2
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Question 14

Tables Used:

- Employees

Find the median salary of all employees.

Example

Output:

median_salary

65000

Question 15

Tables Used:

- Employees
- Departments

Find the difference between the highest and lowest salary in each department.

Example

Output:

department_name	salary_gap
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HR	40000
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Question 16

Tables Used:

- Customers
- Orders

Find customers who placed orders in three consecutive months.

Example

Output:

customer_name

Ayesha

Question 17 *

Tables Used:

- Employees
- Departments

Retrieve the top 3 distinct salaries from each department.

Example

Output:

department_name	salary
IT	120000
IT	110000
IT	100000

HR	90000
HR	85000
HR	80000

Question 18

Tables Used:

- Orders
- OrderDetails
- Products

Find pairs of products that were ordered together more than 10 times.

Example

Output:

product1 product2

Mouse Keyboard

Question 19

Tables Used:

- Logins

Find the longest login streak for each user.

Example

Output:

user_id	longest_streak
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1	5
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Question 20

Tables Used:

- Employees

Find all employees sharing the same salary within the same department.

Example**Output:**

employee_name	salary
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Ali	80000
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Ahmed	80000
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Question 21

Tables Used:

- Orders

Calculate month-over-month sales growth percentage.

Example

Output:

month	growth_percentage
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February	12.5
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Question 22

Tables Used:

- Customers

Find missing customer IDs in a sequential customer database.

Example

Input:

customer_id

1

2

4

Output:

missing_id

3

Question 23

Tables Used:

- Customers
- Orders
- Products
- OrderDetails

Find the highest spending customer in each product category.

Example

Output:

category	customer_name
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Electronics	Sara
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Question 24

Tables Used:

- Employees
- Departments

Find the department having the highest average salary, excluding departments with fewer than 3 employees.

Example

Output:

department_name	average_salary
IT	98000

Question 25

Tables Used:

- Orders

Identify customers who placed more than 5 orders within any 10-minute interval.

Example

Output:

customer_id
